

Hankel Determinants of Geometric Moments and Rational Lambert Values

Erdős Problem #1049

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ABSTRACT

We determine the fixed-base asymptotic of Zudilin's normalised Hankel determinants, including the constant and the factor $N^{-8F(1/q)}$, where $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$. A general theorem for positive geometric moments separates the contribution of the node geometry from the product of the weights; two finite convolutions determine the latter. Separately, rational specialisation of Zudilin's 2004 polynomial forms gives an irrationality region for F larger than that supplied by Bundschuh and Väänänen's criterion. It includes every positive integral power of $31/4$, but excludes $3/2$.

1 Introduction

For each fixed real $q \in (0, 1)$, we determine the asymptotic size of the Hankel determinants arising from Zudilin's q -logarithm approximations. Write $(z; q)_m = \prod_{j=0}^{m-1} (1 - zq^j)$ and $F(t) = \sum_{n \geq 1} (t^n - 1)^{-1}$ for $t > 1$. At the auxiliary parameters $x = z = 1$, the normalisation in [5, (6), §4] gives

$$v_m^*(q) = \sum_{t \geq 0} q^{(m+1)t} \frac{(q; q)_m^3 (q^{t+1}; q)_m}{(q^{m+1+t}; q)_{m+1}}, \quad V_N^*(q) = \det(v_{i+j}^*(q))_{0 \leq i, j < N}. \quad (1)$$

The sums converge for $0 < q < 1$. They also define formal series, since the t th summand has q -order $(m+1)t$.

Theorem 1.1 (the size of V_N^* at a fixed base). *Fix $0 < q < 1$ and write $P = (q; q)_\infty$, $B_N = N(N-1)(2N-1)/6$ and $C_N = (N!)^2(N+1)!/2^N$. There is a $K(q) > 0$ with*

$$V_N^*(q) \sim K(q) C_N q^{B_N} P^{2N} N^{-8F(1/q)} \quad (N \rightarrow \infty).$$

The constant $K(q)$ is the convergent product in (13). Zudilin proved $\text{ord}_q V_N^* \geq B_N$ and the estimate $|V_N^*| \leq q^{N^3/3} \exp(O_q(N^2))$ [5, Lemma 1 and §4]. Theorem 1.1 determines the factorial, exponential and power factors contained in this estimate. We also

* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

recover the exact first formal term $C_N q^{B_N}$. The formal calculation fixes N and expands at $q = 0$; the asymptotic fixes q and lets N grow. Neither limit can be substituted for the other: multiplication by $(1 - q)^{N^3}$ preserves the first formal term but changes the fixed-base logarithm by a cubic quantity. No uniformity as $q \rightarrow 1$ is asserted.

The proof separates the node geometry from the weights. We write $v_m^* = \sum_{k \geq 0} a_k q^{(m+1)k}$ with $a_k > 0$, so that the representing measure has mass $a_k q^k$ at q^k . Heine's identity expresses the determinant as a positive sum over choices of N nodes, with a squared Vandermonde for each choice [7, §2, (2)–(5)]. The first N nodes give a natural normalisation, not an asymptotically complete answer: moving just the last node leaves a contribution of comparable size. We encode the displacements by partitions. A fixed partition moves only finitely many high-index nodes, where the weight ratios tend to 1. A summable bound independent of N controls the sum over all partitions. After division by the first-node contribution, the limit therefore agrees with the constant-weight case, which Cauchy's determinant formula evaluates exactly.

The remaining calculation concerns the weights. Their factorisation involves two-fold and three-fold convolutions of $1/(q; q)_k$, with first relative corrections $-2F(1/q)/(k+1)$ and $-6F(1/q)/(k+2)$. Taking logarithms turns these corrections into a harmonic sum, giving $N^{-8F(1/q)}$; the summable errors affect only the constant. Theorem 2.1 isolates the determinant argument from this coefficient calculation; it also evaluates the determinant with entries $(1 - q^{i+j+1})^{-2}$.

Chowla's conjecture, recorded by Erdős [1, p. 102], asks whether $F(t)$ is irrational for every rational $t > 1$. This is a related arithmetic question, and the fixed-base asymptotic alone does not answer it. In Section 4 we instead specialise the polynomial forms of Zudilin's 2004 construction [4, §5]. They prove irrationality for coprime $a > b \geq 1$ satisfying

$$\frac{\log b}{\log a} < \theta^* = 0.4056830213840605 \dots,$$

where (15) defines θ^* exactly. The constants are inherited from Zudilin; the degree calculation after polynomial cancellation supplies the rational-base estimate. The resulting region includes every positive integral power of $31/4$, which lies outside the earlier sufficient region of Bundschuh and Väänänen [2, Theorem 2]. Neither argument here settles $3/2$.

Section 2 proves the general determinant formula. Section 3 constructs the weights and proves Theorem 1.1. The rational-base argument is self-contained in Section 4, with longer calculations and complementary questions linked in Section 5. All logarithms are natural, empty products and determinants equal 1, and constants in $O_q(\cdot)$ may depend on the fixed q .

2 Geometric moment determinants

For constant weights, the moments are geometric series: $M_m = \sum_{k \geq 0} q^{(m+1)k} = (1 - q^{m+1})^{-1}$. Cauchy's determinant formula evaluates their determinant exactly. Writing

$$P = (q; q)_\infty, \quad M(q) = \prod_{d \geq 1} (1 - q^d)^{-d}, \quad B_N = \sum_{j < N} j^2,$$

we obtain

$$D_N^{(0)} := \det \left(\frac{1}{1 - q^{i+j+1}} \right)_{0 \leq i, j < N} = \frac{q^{B_N} \Delta_N}{\prod_{0 \leq i, j < N} (1 - q^{i+j+1})}, \quad \Delta_N = \prod_{d=1}^{N-1} (1 - q^d)^{2(N-d)}. \quad (2)$$

For each fixed d , the multiplicity of $1 - q^d$ in the denominator increases to d , giving the factor $\mathcal{M}(q)$ in the limit. The numerator supplies two more copies:

$$\frac{\Delta_N}{p^{2N}} = \prod_{d < N} (1 - q^d)^{-2d} \prod_{d \geq N} (1 - q^d)^{-2N} \rightarrow \mathcal{M}(q)^2.$$

Indeed, $\sum d |\log(1 - q^d)| < \infty$, and the logarithm of the second product is $O_q(Nq^N)$. Thus

$$D_N^{(0)} \sim \mathcal{M}(q)^3 q^{B_N} p^{2N}. \quad (3)$$

Under the following hypotheses, replacing the constant weights by a_k multiplies the asymptotic by $\prod_{k < N} a_k$. The ratio limit controls each fixed displacement of a node, while the polynomial bound controls the sum over all displacements.

Theorem 2.1 (geometric moment determinants). *Let $0 < q < 1$ and $a_k > 0$ for $k \geq 0$. Suppose that, for fixed constants $C > 0$ and $\kappa \geq 0$,*

$$\frac{a_{k+h}}{a_k} \rightarrow 1 \quad (k \rightarrow \infty) \text{ for each fixed } h \geq 0, \quad \frac{a_{k+h}}{a_k} \leq C(1+h)^\kappa \quad (k, h \geq 0). \quad (4)$$

For $M_m = \sum_{k \geq 0} a_k q^{(m+1)k}$ and $D_N = \det(M_{i+j})_{0 \leq i, j < N}$, we have

$$D_N \sim \mathcal{M}(q)^3 q^{B_N} p^{2N} \prod_{k=0}^{N-1} a_k \quad (N \rightarrow \infty). \quad (5)$$

For every real s , the weights $a_k = (k+1)^s$ satisfy (4) with $C = 1$ and $\kappa = \max(s, 0)$, and their first N terms have product $(N!)^s$. Taking $s = 1$ and summing the moments gives

$$\det \left(\frac{1}{(1 - q^{i+j+1})^2} \right)_{0 \leq i, j < N} \sim \mathcal{M}(q)^3 N! q^{B_N} p^{2N}. \quad (6)$$

Proof of Theorem 2.1. The bound $a_h \leq Ca_0(1+h)^\kappa$ makes $\sum_{k \geq 0} a_k q^k \delta_{q^k}$ a finite positive measure with moments M_m . Truncate it to finitely many atoms and apply Cauchy–Binet. As the truncation grows, the matrix entries converge, hence so does the determinant; the tuple sum increases because its terms are nonnegative. We obtain

$$D_N = \sum_{0 \leq k_0 < \dots < k_{N-1}} \prod_{i < N} a_{k_i} q^{k_i} \prod_{i < j < N} (q^{k_i} - q^{k_j})^2. \quad (7)$$

The tuple $k_i = i$ contributes $q^{B_N} \Delta_N \prod_{i < N} a_i$. It does not account for the whole asymptotic. For example, replace $(0, 1, \dots, N-2, N-1)$ by $(0, 1, \dots, N-2, N)$. The ratio of its contribution to that of the original tuple is

$$q \frac{a_N}{a_{N-1}} \prod_{d=1}^{N-1} \left(\frac{1 - q^{d+1}}{1 - q^d} \right)^2 = q \frac{a_N}{a_{N-1}} \left(\frac{1 - q^N}{1 - q} \right)^2 \rightarrow \frac{q}{(1 - q)^2}. \quad (8)$$

Only pairs involving the moved node change, so the product telescopes. At $q = 1/2$ the limiting ratio is 2: the displaced tuple contributes twice as much asymptotically as $(0, 1, \dots, N-1)$.

Subtracting $(0, 1, \dots, N-1)$ gives the shifts $\lambda_i = k_i - i$. Since $k_{i+1} \geq k_i + 1$, these are nonnegative and weakly increasing; reversing their order gives a partition. For $N = 4$, for example,

$$\underbrace{(0, 1, 3, 5)}_{(k_i)} \mapsto \underbrace{(0, 0, 1, 2)}_{(\lambda_i)} \mapsto \underbrace{(2, 1, 0, 0)}_{(\mu_j)}.$$

In general, put $\mu_j = \lambda_{N-j}$ for $1 \leq j \leq N$, so that $\mu_1 \geq \dots \geq \mu_N \geq 0$. Division by the contribution of $k_i = i$ turns the summand for this tuple into

$$W_N(\mu) = q^{\sum_{j \leq N} (2j-1)\mu_j} \prod_{j \leq N} \frac{a_{N-j+\mu_j}}{a_{N-j}} \cdot \prod_{1 \leq j < k \leq N} \left(\frac{1 - q^{k-j+\mu_j-\mu_k}}{1 - q^{k-j}} \right)^2. \quad (9)$$

Here “from the right” refers to the increasing indices k_i ; the nodes q^{k_i} themselves decrease towards zero. The j th index from the right contributes once through q^{k_i} and twice through each of its $j-1$ pairs with later indices in the squared Vandermonde. Its displacement therefore has exponent $1 + 2(j-1) = 2j-1$.

Set $W_N(\mu) = 0$ when μ has more than N positive parts, so that every sum has the same index set. Fix a partition with ℓ positive parts. It affects only the last ℓ indices of $(0, 1, \dots, N-1)$: for instance, $(2, 1)$ affects the last two for every $N \geq 2$. Those indices tend to infinity, so the weight ratios in (9) tend to 1 by the fixed-shift hypothesis.

The Vandermonde quotient still contains a growing number of factors, but its dependence on N telescopes. The factors with $j < k \leq \ell$ are fixed, and those with $j > \ell$ equal 1. For $N \geq \ell$ and each $j \leq \ell$, the remaining factors satisfy

$$\prod_{k=\ell+1}^N \frac{1 - q^{k-j+\mu_j}}{1 - q^{k-j}} = \frac{(q^{N+1-j}; q)_{\mu_j}}{(q^{\ell+1-j}; q)_{\mu_j}} \rightarrow \frac{1}{(q^{\ell+1-j}; q)_{\mu_j}}. \quad (10)$$

Thus each $W_N(\mu)$ has the same limit as for constant weights.

To pass this limit through the sum over partitions, we need a bound independent of N that remains summable as both the number and size of the parts grow. Only factors with $j \leq \ell$ can differ from 1. For each such j , the denominator product in the Vandermonde quotient is at least P , and its numerator product is at most 1. Together with the polynomial shift bound, this gives

$$W_N(\mu) \leq (CP^{-2})^\ell \prod_{j=1}^{\ell} (1 + \mu_j)^\kappa q^{(2j-1)\mu_j}.$$

Let $A = \sum_{u \geq 1} (1+u)^\kappa q^{u-1} < \infty$. Dropping the ordering of the positive parts, we obtain

$$\begin{aligned} \sum_{\ell(\mu)=\ell} \sup_N W_N(\mu) &\leq (CP^{-2})^\ell \prod_{j=1}^\ell \sum_{u \geq 1} (1+u)^\kappa q^{(2j-1)u} \\ &\leq (CP^{-2}A)^\ell q^{\ell^2}. \end{aligned}$$

Here $q^{(2j-1)u} \leq q^{2j-1} q^{u-1}$ and $1+3+\dots+(2\ell-1) = \ell^2$. Each positive part pays at least its displacement exponent $2j-1$. Their combined cost q^{ℓ^2} dominates the exponential factor $(CP^{-2}A)^\ell$, so the bound is summable over ℓ , independently of N .

Dominated convergence now shows that $\sum_\mu W_N(\mu)$ has the same limit as for $a_k = 1$. We need not evaluate the limiting partition sum directly: by (2), it is $\mathcal{M}(q)$, the reciprocal Cauchy denominator after division by the principal tuple. The other two copies of $\mathcal{M}(q)$ come from Δ_N/P^{2N} . Restoring the principal tuple thus gives

$$\frac{D_N}{D_N^{(0)} \prod_{k < N} a_k} \rightarrow 1.$$

Combining this limit with (3) proves the theorem. $\square$

3 The weights of Zudilin's moments

We apply Theorem 2.1 by collecting the dependence on m in (1) into the single variable $w = q^{m+1}$. For this purpose, set

$$G_q(w) = \frac{1}{(w; q)_\infty^3} \sum_{t \geq 0} \frac{w^t}{(q; q)_t} \frac{(q^t w^2; q)_\infty}{(q^t w; q)_\infty^2}, \quad a_k(q) = [w^k] P^4 G_q(w). \quad (11)$$

The product identities $(q; q)_m = P/(q^{m+1}; q)_\infty$ and $(q^a; q)_m = (q^a; q)_\infty / (q^{a+m}; q)_\infty$ give

$$v_m^* = P^4 G_q(q^{m+1}) = \sum_{k \geq 0} a_k(q) q^{(m+1)k}. \quad (12)$$

These identities hold formally in $\mathbb{Z}[[q, w]]$ and, for fixed $0 < q < 1$, analytically for $|w| < 1$. In the formal identity only finitely many t contribute to any w -coefficient. For the analytic identity, we use the absence of denominator zeros in that disc and the locally uniform bounds on the product tails. The factor w^t then gives locally uniform convergence of the sum. The product formula for G_q does not yet make the signs of its coefficients evident. We need positive weights, control of their shifted ratios, and enough precision to multiply the first N weights. The finite factorisation below supplies positivity and the ratio bound; its two convolutions then supply the precision.

3.1 Positive weights

For $r \geq 1$ and $k \geq 0$, set

$$R_k^{(r)}(q) = \sum_{n_1 + \dots + n_r = k} \frac{(q; q)_k}{\prod_{j=1}^r (q; q)_{n_j}},$$

over r nonnegative parts. These are the multivariate Rogers–Szegő polynomials at unit arguments, written as sums of Gaussian multinomial polynomials [19]. At $q = 0$, every Gaussian multinomial equals 1, so they count weak compositions (zero parts are allowed), and

$$c_k := R_k^{(2)}(0)R_k^{(3)}(0) = \frac{(k+1)^2(k+2)}{2}, \quad \prod_{k \leq N} c_k = C_N.$$

Proposition 3.1 (the positive moment weights). *For every $k \geq 0$,*

$$a_k = P^4 \frac{R_k^{(2)} R_k^{(3)}}{(q; q)_k}.$$

At fixed $0 < q < 1$ these weights satisfy

$$P^5 c_k \leq a_k \leq P^{-1} c_k, \quad \frac{a_{k+h}}{a_k} \leq P^{-6} (1+h)^3 \quad (k, h \geq 0).$$

Proof. We use the Gaussian binomial coefficients $\begin{bmatrix} n \\ j \end{bmatrix}_q = (q; q)_n / ((q; q)_j (q; q)_{n-j})$ for $0 \leq j \leq n$, and put $\mathcal{E}(z) = (z; q)_\infty^{-1}$. Euler’s expansion gives $\mathcal{E}(z)^r = \sum_{n \geq 0} R_n^{(r)} z^n / (q; q)_n$. The desired factorisation multiplies two coefficients at the same index k , rather than taking a coefficient of $\mathcal{E}^5$. We obtain it by splitting $k = n + j$: the factor $R_k^{(3)}$ stays fixed, and summing the two factorial denominators supplies $R_k^{(2)}$. To produce this split, use the unnormalised q -difference operator $\partial_q f(z) = (f(z) - f(qz))/z$. Since $\partial_q z^j = (1 - q^j)z^{j-1}$, it cancels the last factor of $(q; q)_j$ and shifts the numerator index:

$$\partial_q \mathcal{E} = \mathcal{E}, \quad \partial_q^n \mathcal{E}(z)^3 = \sum_{j \geq 0} \frac{R_{n+j}^{(3)}}{(q; q)_j} z^j.$$

The q -Leibniz rule, obtained by induction from the product rule, is

$$\partial_q^n (fg)(z) = \sum_{j=0}^n \begin{bmatrix} n \\ j \end{bmatrix}_q (\partial_q^j f)(q^{n-j}z) (\partial_q^{n-j} g)(z).$$

Let $h_k(z) = \sum_{j=0}^k \begin{bmatrix} k \\ j \end{bmatrix}_q (z; q)_j$. Applying the rule to $\mathcal{E} \cdot \mathcal{E}$ and using $\mathcal{E}(q^m z) = (z; q)_m \mathcal{E}(z)$, we obtain $\partial_q^k \mathcal{E}(z)^2 = \mathcal{E}(z)^2 h_k(z)$ (for $k = 1$, both sides equal $\mathcal{E}(z)^2(2 - z)$). A second application, this time to $\mathcal{E} \cdot \mathcal{E}^2$, gives

$$\partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(z)^3 \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix}_q (z; q)_k h_k(z),$$

and hence

$$\sum_{n \geq 0} \frac{w^n}{(q; q)_n} \partial_q^n \mathcal{E}(z)^3 = \mathcal{E}(w) \mathcal{E}(z)^3 \sum_{k \geq 0} \frac{w^k (z; q)_k}{(q; q)_k} h_k(z).$$

The left-hand side is $\sum_{n, j \geq 0} R_{n+j}^{(3)} w^{n+j} / ((q; q)_n (q; q)_j)$. Putting $z = w$ collects all pairs with the same sum $n + j$. The coefficient of w^k is consequently

$R_k^{(3)} \sum_{j \leq k} ((q; q)_j (q; q)_{k-j})^{-1} = R_k^{(2)} R_k^{(3)} / (q; q)_k$. On the right-hand side, put $z = w$, expand h_k , and write $k = j + t$. The identity $(w; q)_{j+t} = (w; q)_j (wq^j; q)_t$ gives

$$\mathcal{E}(w)^4 \sum_{j \geq 0} \frac{w^j (w; q)_j^2}{(q; q)_j} \sum_{t \geq 0} \frac{(wq^j; q)_t}{(q; q)_t} w^t.$$

The inner sum is evaluated by the q -binomial theorem:

$$\sum_{t \geq 0} \frac{(wq^j; q)_t}{(q; q)_t} w^t = \frac{(w^2 q^j; q)_\infty}{(w; q)_\infty}.$$

Finally, $(w; q)_j = (w; q)_\infty / (wq^j; q)_\infty$ cancels two of the five factors $\mathcal{E}(w)$, leaving

$$\mathcal{E}(w)^3 \sum_{j \geq 0} \frac{w^j}{(q; q)_j} \frac{(w^2 q^j; q)_\infty}{(wq^j; q)_\infty^2} = G_q(w).$$

We may read each identity coefficientwise, or as an absolutely convergent expansion when $|z|$ and $|w|$ are sufficiently small.

For the bounds, we write $b_k^{(r)} = [z^k](z; q)_\infty^{-r}$, so that $R_k^{(r)} = (q; q)_k b_k^{(r)}$ and $a_k = P^4 (q; q)_k b_k^{(2)} b_k^{(3)}$. By Euler's expansion, $b_k^{(r)}$ is an r -fold convolution of $1/(q; q)_n$, whose terms lie in $[1, P^{-1}]$. There are $\binom{k+r-1}{r-1}$ compositions of k into r nonnegative parts, and we deduce

$$\binom{k+r-1}{r-1} \leq b_k^{(r)} \leq P^{-r} \binom{k+r-1}{r-1}.$$

Combining these at $r = 2, 3$ with $P \leq (q; q)_k \leq 1$ and $c_k = (k+1) \binom{k+2}{2}$ gives $P^5 c_k \leq a_k \leq P^{-1} c_k$. The shift bound follows from $c_{k+h}/c_k \leq (1+h)^3$. $\square$

The positivity here is $a_k(q) > 0$ at each fixed real $q \in (0, 1)$, not coefficientwise positivity of its formal q -series: already $a_0(q) = P^4 = 1 - 4q + O(q^2)$. The bounds prove the uniform shift estimate, but replacing a_k by c_k in a product of N terms requires greater accuracy. A relative error of order $1/k$ accumulates logarithmically; an error of order $1/k^2$ is summable. We therefore compute a_k/c_k to first order. This also gives the fixed-shift limit.

3.2 The coefficient asymptotic

Proof of Theorem 1.1. Put $L = F(1/q)$, $e_k = (q; q)_k^{-1}$ and $E = P^{-1}$. Replacing every e_k by E in the two convolution sums gives $E^2(k+1)$ and $E^3 \binom{k+2}{2}$. The factors $k+1$ and $\binom{k+2}{2}$ count nonnegative pairs and triples with sum k . For a fixed i , one pair and $k-i+1$ triples have first coordinate i . Consequently, the summable differences $E - e_i$ produce a constant correction in the two-fold convolution and a linear correction in the three-fold convolution. The terms containing two or more such differences contribute to the error bounds below.

Writing $d_k = E - e_k$, the identity $e_{k+1} - e_k = q^{k+1}e_{k+1}$ gives

$$d_k = \sum_{j>k} q^j e_j, \quad 0 \leq d_k \leq \frac{Eq^{k+1}}{1-q}.$$

Thus $\sum d_k$ and $\sum kd_k$ converge. To evaluate the former, use Euler's identity $\mathcal{E}(z) = \sum_{k \geq 0} e_k z^k = (z; q)_\infty^{-1}$ for $|z| < 1$. Tonelli's theorem and logarithmic differentiation at $z = q$ give

$$S := \sum_{k \geq 0} d_k = \sum_{j \geq 1} jq^j e_j = q\mathcal{E}'(q) = E \sum_{r \geq 1} \frac{q^r}{1-q^r} = EL.$$

In the double sum, $q^j e_j$ occurs once for each $k = 0, \dots, j-1$, which explains the factor j . Logarithmic differentiation is justified by locally uniform convergence on $|z| < 1$.

Let $b_k^{(r)} = [z^k] \mathcal{E}(z)^r$. The finite-sum formula in Proposition 3.1 becomes

$$a_k = P^4(q; q)_k b_k^{(2)} b_k^{(3)}.$$

We expand each finite convolution using $e_k = E - d_k$. For two factors,

$$b_k^{(2)} = E^2(k+1) - 2E \sum_{i=0}^k d_i + \sum_{i=0}^k d_i d_{k-i} = E^2(k+1-2L) + O_q((k+1)q^k).$$

For three factors, the terms with one d_i contribute $-3E^2 \sum_{i=0}^k (k-i+1)d_i$. Here the sum equals $(k+1)S + O_q(1)$ because $\sum id_i < \infty$. Terms with two d_i are bounded by $3ES^2$, and the term with three is $O_q((k+1)^2 q^k)$. It follows that

$$b_k^{(3)} = E^3 \left(\binom{k+2}{2} - 3L(k+1) \right) + O_q(1).$$

Dividing the correction $-3L(k+1)$ by the leading term $\binom{k+2}{2}$ gives the relative correction $-6L/(k+2)$. The two-fold convolution contributes $-2L/(k+1)$ in the same way. Since $(q; q)_k = P(1 + O_q(q^k))$, the leading constants in $a_k = P^4(q; q)_k b_k^{(2)} b_k^{(3)}$ multiply to $P^5 E^5 = 1$. Replacing $(k+2)^{-1}$ by $(k+1)^{-1}$ changes only the $O_q((k+1)^{-2})$ error. Thus the coefficients $2L$ and $6L$ add, and with $c_k = (k+1)^2(k+2)/2$ we obtain

$$\begin{aligned} \frac{a_k}{c_k} &= \left(1 - \frac{2L}{k+1} + O_q(q^k) \right) \left(1 - \frac{6L}{k+2} + O_q((k+1)^{-2}) \right) (1 + O_q(q^k)) \\ &= 1 - \frac{8L}{k+1} + O_q((k+1)^{-2}). \end{aligned}$$

In particular $a_{k+1}/a_k \rightarrow 1$, and multiplication of finitely many adjacent ratios gives $a_{k+h}/a_k \rightarrow 1$ for each fixed h . The uniform shift bound follows from Proposition 3.1, so Theorem 2.1 applies.

Let γ_E denote Euler's constant and put

$$\mathcal{A}(q) = e^{-8\gamma_E L} \prod_{k \geq 0} \left(\frac{a_k}{c_k} e^{8L/(k+1)} \right), \quad K(q) = \mathcal{A}(q) M(q)^3. \quad (13)$$

For sufficiently large k , $\log(1 + u) = u + O(u^2)$ gives $\log(a_k/c_k) + 8L/(k + 1) = O_q((k + 1)^{-2})$. This summable tail makes the product converge to a nonzero limit; the finitely many initial factors are positive by Proposition 3.1. Hence $0 < A(q) < \infty$. Using $\prod_{k < N} c_k = C_N$, separate the harmonic sum from this convergent remainder:

$$\log \frac{\prod_{k < N} a_k}{C_N} = -8L \sum_{k < N} \frac{1}{k + 1} + \sum_{k < N} \left(\log \frac{a_k}{c_k} + \frac{8L}{k + 1} \right).$$

The second sum tends to $\log A(q) + 8\gamma_E L$. Using $\sum_{k < N} (k + 1)^{-1} = \log N + \gamma_E + o(1)$, the Euler-constant terms cancel, giving $\prod_{k < N} a_k \sim A(q) C_N N^{-8L}$. Substitution in (5) proves the theorem, or equivalently

$$\log V_N^* = B_N \log q + \log C_N + 2N \log P - 8L \log N + \log K(q) + o(1).$$

□

The logarithmic formula separates the scales: $B_N \log q$ is cubic in N , $\log C_N$ has order $N \log N$, and the remaining terms are linear, logarithmic and constant. The cubic weights give C_N ; their first relative correction gives $N^{-8F(1/q)}$. That correction refines the size without changing the cubic rate.

3.3 The first nonzero formal term

The order of a nonzero formal series is the least exponent with nonzero coefficient. We now fix N and determine the first term in the formal q -expansion. The shifted term in (8) then has an extra factor of q . More generally, every nonempty partition has positive displacement exponent. Thus a single tuple determines the first formal term, although all partitions were needed for the fixed-base constant.

Theorem 3.2 (the first nonzero term of the Hankel determinant). *For every $N \geq 1$,*

$$\text{ord}_q V_N^* = \frac{N(N-1)(2N-1)}{6},$$

and the coefficient of the first nonzero monomial is

$$[q^{N(N-1)(2N-1)/6}] V_N^* = \frac{(N!)^2 (N+1)!}{2^N}.$$

Proof. At $q = 0$, the $t = 0$ term of $G_q(w)$ is $(1 - w^2)/(1 - w)^5$, while the remaining terms sum to $w/(1 - w)^4$. Thus

$$G_0(w) = \frac{1 + 2w}{(1 - w)^4}, \quad a_k(0) = c_k > 0.$$

We compute modulo q^{M+1} by truncating each moment sum in (12) at $k = M$. Since all omitted terms are divisible by q^{M+1} , multilinearity leaves the determinant unchanged modulo that power. Applying finite Cauchy–Binet and then allowing M to increase, we obtain the coefficientwise identity

$$V_N^* = \sum_{k_0 < \dots < k_{N-1}} \left(\prod_{i < N} a_{k_i}(q) q^{k_i} \right) \prod_{i < j < N} (q^{k_i} - q^{k_j})^2.$$

The order of the summand indexed by (k_i) is $\sum_{i \leq N} (2N - 1 - 2i)k_i$. Because $k_i \geq i$ and all the weights $2N - 1 - 2i$ are positive, we obtain the unique minimum at $k_i = i$, with order and coefficient

$$\sum_{i \leq N} (2N - 1 - 2i)i = B_N, \quad \prod_{i \leq N} c_i = C_N.$$

No other tuple has this order, so no coefficient from another tuple can cancel it. This argument needs the positive constant terms $a_k(0) = c_k$, not nonnegativity of all their formal coefficients. For $N = 2$, the pair $(0, 1)$ contributes $6q + O(q^2)$, and every other increasing pair has order at least two. $\square$

This proves equality in Zudilin's bound [5, §4, pp. 6–7]. An alternative argument, which computes the first nonzero term of every transformed row, is given in [the companion paper](#).

4 Rational bases

We now use the 2004 coefficient family. If $U, V \in \mathbb{Z}[X]$ have degree at most W , evaluation at a/b requires the factor b^W to produce integer coefficients. A common polynomial factor should therefore be cancelled first. For example, at $3/2$ the pair $(X+1)(X+2), (X+1)(X+3)$ gives $(35, 45)$ after clearing to degree 2. After cancelling $X+1$, clearing to degree 1 gives $(7, 9)$. For Zudilin's forms, the degrees are of quadratic order in the index n . Write K_n for the degree of the original coefficient and W_n for its degree after cancellation. The cancelled remainder has leading size $x^{-(K_n - W_n)}$ at a fixed real base $x > 1$. At $x = a/b$, clearing the coefficients therefore has the balance

$$b^{W_n} (a/b)^{-(K_n - W_n)} = \frac{b^{K_n}}{a^{K_n - W_n}}. \quad (14)$$

The proof below shows that the remaining factors contribute only $o(n^2)$ to the logarithm. Hence the limits $K_n/n^2 \rightarrow C_1$ and $(K_n - W_n)/n^2 \rightarrow C_0$ give the sufficient inequality $C_1 \log b < C_0 \log a$. We first justify cancellation in $\mathbb{Z}[X]$ and calculate the degrees, before specialising X to a/b .

We use the parameter direction from Zudilin's construction [4, §5, pp. 161–162], together with its thirteen intervals and constants C_1, C_0 . At integer bases his theorem gives the irrationality-exponent bound $C_1/C_0 = 2.46497868\dots$ [4, Thm. 1, p. 154]. Let $\psi_1(u) = \sum_{k \geq 0} (k+u)^{-2}$ for $u > 0$, and let $\mathcal{I}$ consist of the thirteen intervals listed in the proof. We put

$$C_1 = \frac{1091}{2}, \quad J = \sum_{[u,v] \in \mathcal{I}} (\psi_1(u) - \psi_1(v)), \quad C_0 = 266 - \frac{3}{\pi^2} (225 - J). \quad (15)$$

The intervals are disjoint and lie in $[1/14, 1)$, so $0 \leq J \leq \psi_1(1/14) - \psi_1(1) < 196$. Together with $\pi > 3$, these bounds give $0 < C_0 < 266 < C_1/2$. Thus $\theta^* = C_0/C_1$ and $\mu = C_1/C_0$ are positive reciprocal constants. The notation $\mu_{\text{irr}}(\xi)$ instead denotes the irrationality exponent of a value. The estimates below give a sufficient cutoff; its optimality is unknown.

Theorem 4.1 (rational-base region for Zudilin's forms). *Let $a > b \geq 1$ be coprime integers with*

$$b^\mu < a, \quad \text{equivalently} \quad \frac{\log b}{\log a} < \theta^*,$$

$$\theta^* = \frac{C_0}{C_1} = 0.4056830213840605 \dots,$$

$$\mu = \frac{C_1}{C_0} = 2.4649786835749750 \dots$$

Then $F(a/b)$ is irrational.

The condition holds for every integer base, since $b = 1$ gives $\log b / \log a = 0$. For a fixed denominator $b > 1$, it requires $a > b^{2.46497868\dots}$, much more than $a > b$; nevertheless it gives infinitely many reduced noninteger rational bases for each such b . The base $31/4$ satisfies the condition, and taking a common positive integer power of the numerator and denominator leaves their logarithmic ratio unchanged. Since $\theta^* < 1/2 < \log 2 / \log 3$, the condition fails at $3/2$. At equality $\log b / \log a = \theta^*$ the quadratic exponent vanishes, so the estimates below give no conclusion.

Proof of Theorem 4.1. We first identify the integral polynomials obtained by cancellation. We then compute their degrees and the size of the cleared remainder, which will justify (14) to quadratic order.

The source forms and polynomial cancellation. Fix $n \geq 1$ and set

$$a_0 = 14n + 1, \quad a_1 = 12n + 1, \quad a_2 = 14n + 1, \quad \beta = 27n + 2, \quad N = 15n.$$

In this proof N denotes the cyclotomic cutoff. We use the coefficient pair A_n, B_n of the source identities [4, (8)–(11), pp. 156–157], with $H_n = A_n F - B_n$. Let Φ_ℓ be the cyclotomic polynomial whose roots are the primitive ℓ th roots of unity. Put $D_N(X) = \prod_{\ell=1}^N \Phi_\ell(X)$, $M_n = 266n^2 + 34n + 1$, and

$$\Omega_n(X) = \prod_{\ell=2}^N \Phi_\ell(X)^{\nu_\ell}, \quad \nu_\ell = \omega(n/\ell),$$

where the periodic function

$$\omega(x) = \max\{0, \lfloor 14x \rfloor + \lfloor 13x \rfloor - \lfloor 12x \rfloor - \lfloor 15x \rfloor, \\ 2\lfloor 14x \rfloor - \lfloor 13x \rfloor - \lfloor 15x \rfloor\}$$

has period 1: the coefficients of the floor arguments in each difference have equal sums, $14 + 13 = 12 + 15$ and $2 \cdot 14 = 13 + 15$. On $[0, 1)$ it is zero or one, with support

$$\begin{aligned} &[1/14, 1/12), [1/7, 1/6), [3/14, 1/4), [2/7, 1/3), \\ &[5/14, 2/5), [3/7, 7/15), [1/2, 8/15), [4/7, 3/5), \\ &[9/14, 2/3), [5/7, 11/15), [11/14, 4/5), \\ &[6/7, 13/15), [13/14, 14/15). \end{aligned}$$

These are the intervals $\mathcal{I}$ used to define J . The function ω , these thirteen intervals and the exponents $\nu_\ell = \omega(n/\ell)$ of the source's (22) are printed at [4, pp. 161–162].

We first bound the remainder before cancellation. For $x > 1$ we put $q = 1/x$ and use identity (9) on p. 156 of the source, which gives the positive representation

$$H_n(x) = \sum_{t \geq 0} q^{a_0 t} \frac{(q^{t+1}; q)_{a_1-1}}{(q; q)_{a_1-1}} \frac{(q; q)_{\beta-a_2-1}}{(q^{a_2+t}; q)_{\beta-a_2}}.$$

The denominator length is $\beta - a_2$, as in the gamma expression (7) and residues (8) of [4, p. 156]; the unnumbered display of $R(T)$ on that page has length $\beta - a_2 - 1$ instead. Each of the four finite products displayed above lies between $P = (q; q)_\infty > 0$ and 1, so

$$P^2 \leq H_n(x) \leq \frac{P^{-2}}{1 - q^{a_0}}.$$

In particular $H_n(x) > 0$ and $\log H_n(x) = O_x(1)$.

To identify the integral polynomials we use the inclusion in Zudilin's Lemma 7, display (23) [4, p. 161]. The proof compares the cyclotomic valuations of the coefficient polynomials under six admissible parameter permutations, and gives the divisibility in $\mathbb{Z}[X]$ that we need before evaluation. Its parameter vector is $n(13, 14, 12, 14, 15, 13)$, its maximum is $15n$, and $\beta - a_1 - a_2 = n > 0$, which is the positivity condition $s > 0$ of the lemma. The conditions (14) of [4, p. 157], $a_1 \leq a_2$ and $a_1 + a_2 \leq \beta \leq a_0 + a_2$, hold as well, so the exponent in (23) is the integer (16), which equals M_n . Lemma 7 therefore gives, for every $n \geq 1$,

$$\Lambda_n(X) = X^{-M_n} \frac{D_N(X)}{\Omega_n(X)} H_n(X) = U_n(X)F(X) - V_n(X), \quad U_n, V_n \in \mathbb{Z}[X].$$

This is the polynomial conclusion before integer specialisation in the source's (24), p. 162. The coefficients are the cancelled source coefficients:

$$U_n = X^{-M_n} (D_N / \Omega_n) A_n, \quad V_n = X^{-M_n} (D_N / \Omega_n) B_n.$$

Indeed, coefficients of F and 1 over $\mathbb{Q}(X)$ are unique because F is not a rational function. To see this, we split $F(e^h)$ at $m = 1/h$ as $h \downarrow 0$. For $m \leq 1/h$, the inequalities $y^{-1} - 1 \leq (e^y - 1)^{-1} \leq y^{-1}$ for $0 < y \leq 1$ give $h^{-1} \sum_{m \leq 1/h} m^{-1} + O(h^{-1})$. Bounding the remaining terms by a geometric tail gives a further $O(h^{-1})$, and therefore $F(e^h) = h^{-1} \log(1/h) + O(h^{-1})$, which has no rational-function pole order at $X = 1$. This establishes uniqueness of the coefficient functions; it assumes nothing about an individual value $F(a/b)$. The cancelled remainder is positive for $x > 1$, since D_N / Ω_n is a product of cyclotomic polynomials positive there.

The degrees after cancellation. The Gaussian binomial polynomial $\begin{bmatrix} m \\ k \end{bmatrix}_X = \prod_{j=1}^k (1 - X^{m-k+j}) / (1 - X^j)$ is monic of degree $k(m - k)$ for integers $0 \leq k \leq m$. The source's (8) and (10) on p. 156 give the coefficient whose degree we need:

$$A_n(X) = \sum_{k=a_2}^{\beta-1} (-1)^{a_1+a_2+k+1} X^{a_0 k + \binom{a_1}{2} - \binom{\beta-a_2}{2} + \binom{\beta-k}{2}} \begin{bmatrix} k-1 \\ a_1-1 \end{bmatrix}_X \begin{bmatrix} \beta-a_2-1 \\ \beta-k-1 \end{bmatrix}_X.$$

Let $d_{n,k}$ be the degree of the k th summand. Adding the monomial exponent and the degrees of the two Gaussian factors gives

$$\begin{aligned} d_{n,k} &= a_0 k + \binom{a_1}{2} - \binom{\beta - a_2}{2} + \binom{\beta - k}{2} \\ &\quad + (a_1 - 1)(k - a_1) + (\beta - k - 1)(k - a_2) \\ &= \frac{-k^2 + 80kn + 3k - 340n^2 - 26n}{2}. \end{aligned}$$

Hence $d_{n,k+1} - d_{n,k} = 40n + 1 - k > 0$ for $a_2 \leq k \leq \beta - 2 = 27n$. The last summand alone therefore determines the leading term, giving

$$K_n := \deg A_n = \frac{1091n^2 + 81n + 2}{2}, \quad W_n := \deg U_n = K_n - M_n + \sum_{\ell \leq 15n} (1 - \nu_\ell) \varphi(\ell).$$

Here $\varphi(\ell) = \deg \Phi_\ell$ is Euler's totient function, and $\nu_1 = \omega(n) = 0$. In particular,

$$K_n - W_n = M_n - \deg(D_N/\Omega_n).$$

Thus $K_n - W_n$ measures the net degree removed by the normalisation: the monomial removes M_n , and the surviving cyclotomic product restores $\deg(D_N/\Omega_n)$. This same net degree governs the remainder decay.

The bounds for H_n are uniform for $x \geq 2$: then $P \geq \prod_{j \geq 1} (1 - 2^{-j}) > 0$ and $(1 - x^{-a_0})^{-1} \leq 2$. For fixed n , therefore, $H_n(x) = O(1)$ and $\Lambda_n(x) = O(x^{W_n - K_n})$ as $x \rightarrow \infty$. The unique top summand of A_n has leading coefficient $(-1)^{a_1 + a_2 + \beta} = (-1)^n$. Since the Gaussian factors and D_N/Ω_n are monic, U_n has that leading coefficient too. Moreover $F(x) = x^{-1} + O(x^{-2})$ and $K_n \geq 2$, so

$$V_n(x) = U_n(x)F(x) - \Lambda_n(x) = (-1)^n x^{W_n - 1} + O(x^{W_n - 2}).$$

Here $W_n \geq K_n - M_n = (559n^2 + 13n)/2 > 0$. Thus $\deg V_n = W_n - 1$, with the same unit leading coefficient as U_n . For every prime $p \mid b$, reduction of the cleared first coordinate gives

$$b^{W_n} U_n(a/b) \equiv (-1)^n a^{W_n} \not\equiv 0 \pmod{p}.$$

Thus $U_n(a/b)$ has reduced denominator exactly b^{W_n} , the least common clearing denominator for this pair. Every common integer divisor of this cleared row is consequently coprime to b . Thus b^{W_n} is an exact clearing cost for the chosen pair, not a lower bound for every possible approximation: common factors at other primes and combinations of different rows remain separate questions.

The quadratic degree limits. The degree calculation kept n fixed and let $x \rightarrow \infty$. For the irrationality argument we now keep x fixed and let $n \rightarrow \infty$. We first find the quadratic part of the degree removed by cancellation. The limits below are the cyclotomic limits of [4, Lemmas 1–2, p. 155]. The proof uses the argument through reciprocal intervals, the summatory totient estimate and the trigamma function from the proof of [3, Lemma 1, p. 466], with an explicit truncation of the block sum. The elementary summatory estimate $\sum_{\ell \leq y} \varphi(\ell) = 3y^2/\pi^2 + O(y \log y)$ gives

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \varphi(\ell) \rightarrow \frac{675}{\pi^2}.$$

For $[u, v) \in \mathcal{I}$, the condition $\{n/\ell\} \in [u, v)$ is the disjoint union of intervals $n/(k+v) < \ell \leq n/(k+u)$ for $k \geq 0$. We may apply the same summatory estimate term by term to any finite set of k . If we stop before $k = K$, with $K \geq 1$, every omitted block has $\ell \leq n/(K+u)$. Thus the elementary bound $\sum_{\ell \leq y} \varphi(\ell) \leq y^2$ for $y \geq 0$ gives the uniform estimate

$$\frac{1}{n^2} \sum_{\ell \leq n/(K+u)} \varphi(\ell) \leq \frac{1}{(K+u)^2}.$$

Indeed, for $y \geq 1$ one can bound $\varphi(\ell)$ by ℓ and sum; for $0 \leq y < 1$ the sum is empty. This bound is uniform in n . We can now let n and then K tend to infinity, obtaining

$$\frac{1}{n^2} \sum_{\ell \leq 15n} \nu_\ell \varphi(\ell) \rightarrow \frac{3}{\pi^2} \sum_{[u,v) \in \mathcal{I}} \sum_{k \geq 0} \left(\frac{1}{(k+u)^2} - \frac{1}{(k+v)^2} \right) = \frac{3J}{\pi^2}.$$

Here no contributing index exceeds $14n$, since $u \geq 1/14$. Consequently

$$K_n/n^2 \rightarrow C_1, \quad (K_n - W_n)/n^2 \rightarrow C_0.$$

The cleared remainder. We return to a fixed real base $x > 1$ and estimate the cyclotomic product. Writing μ_{Mob} for the Möbius function, we have

$$\log \Phi_\ell(x) - \varphi(\ell) \log x = \sum_{d|\ell} \mu_{\text{Mob}}(d) \log(1 - x^{-\ell/d}).$$

The total absolute error over $\ell \leq 15n$ is $O_x(n)$, since it is at most

$$15n \sum_{d \geq 1} \frac{-\log(1 - x^{-d})}{d}.$$

The series converges because $-\log(1 - x^{-d}) \leq x^{-d}/(1 - x^{-1})$. Since $0 \leq 1 - \nu_\ell \leq 1$, it follows that

$$\log \Lambda_n(x) = -(K_n - W_n) \log x + O_x(n).$$

We can now clear the denominators at the rational base. Since $U_n, V_n \in \mathbb{Z}[X]$ and their degrees are at most W_n , the numbers $b^{W_n} U_n(a/b)$ and $b^{W_n} V_n(a/b)$ are integers. Moreover

$$\begin{aligned} \log(b^{W_n} \Lambda_n(a/b)) &= K_n \log b - (K_n - W_n) \log a + O_{a/b}(n) \\ &= (C_1 \log b - C_0 \log a) n^2 + o(n^2). \end{aligned}$$

The coefficient is negative under the theorem's hypothesis, so the cleared forms tend to zero. Their positive-series representation also guarantees nonvanishing; no independence of two successive forms is needed. If $F(a/b) = r/s$ with integers r, s and $s \neq 0$, every nonzero integer-coefficient form has absolute value at least $1/|s|$. The positive forms above eventually violate this bound. $\square$

Corollary 4.2. $F((31/4)^r)$ is irrational for every integer $r \geq 1$.

Proof. The exact inequalities $31^2 < 4^5$ and $4^{200} < 31^{81}$ give $2/5 < \log 4 / \log 31 < 81/200$. The first term of each trigamma difference yields

$$J \geq \sum_{[u,v] \in \mathcal{I}} (u^{-2} - v^{-2}) = \frac{2015640690251}{25971865920}.$$

Using $\pi > 157/50$ gives the rational lower bound

$$\theta^* > \frac{2359630009523263}{5820307922172744} > \frac{81}{200}.$$

Taking a common positive integer power preserves both the logarithmic ratio and coprimality, so the same criterion applies for every $r \geq 1$. $\square$

Corollary 4.3 (an irrationality measure uniform over powers). *For coprime $a > b \geq 1$ with $\theta = \log b / \log a < \theta^*$ and every integer $r \geq 1$,*

$$\mu_{\text{irr}}(F((a/b)^r)) \leq \frac{1 - \theta}{\theta^* - \theta}.$$

Here $\mu_{\text{irr}}(\xi)$ is the supremum of the exponents ν for which $|\xi - p/q| < q^{-\nu}$ has infinitely many reduced rational solutions. In particular, $\mu_{\text{irr}}(F((31/4)^r)) < 301$ for every $r \geq 1$.

Proof. We follow the passage from linear forms to an exponent bound used by Zudilin at integer bases [4, p. 162]. Applying the estimates separately at each fixed base $(a/b)^r$ will give a bound independent of r . We use the polynomials U_n, V_n from Theorem 4.1, with $W_n = \deg U_n$. For irrationality it was enough that the positive remainders tend to zero. To bound the irrationality exponent, we also need a lower bound for those remainders and an upper bound for the coefficients. The logarithmic remainder asymptotic already gives the former; we now prove the latter. In that construction the coefficient of F before cancellation is a sum of $O(n)$ Laurent monomials times two Gaussian binomial polynomials. Each Gaussian polynomial has nonnegative coefficients summing to at most 2^{27n+2} . Thus the sum of the absolute coefficients is $\exp(O(n))$. Its largest exponent is $K_n = (1091n^2 + 81n + 2)/2$, so its absolute value at a fixed $x > 1$ is at most $x^{K_n} \exp(O(n))$. The cyclotomic estimate in the same proof bounds the normalising multiplier by $x^{W_n - K_n} \exp(O_x(n))$. Multiplication gives

$$|U_n(x)| \leq x^{W_n} \exp(O_x(n)) \quad (x > 1 \text{ fixed}).$$

Set $\xi = F(a/b)$, $Q_n = b^{W_n} U_n(a/b)$ and $P_n = b^{W_n} V_n(a/b)$. With

$$\alpha = (C_1 - C_0) \log a, \quad \tau = C_0 \log a - C_1 \log b > 0,$$

we have $\log(Q_n \xi - P_n) = -\tau n^2 + o(n^2)$ and $|Q_n| \leq \exp(\alpha n^2 + o(n^2))$.

In this proof q now denotes an approximation denominator, not the fixed geometric base of Sections 2 and 3. We choose n^2 near $(\log q)/\tau$ so that the remainder is at most $1/(2q)$ while the coefficient $|Q_n|$ remains controlled. The required comparison is elementary. For integers A, B, p, q with $q > 0$, if $L = A\xi - B$ and $2q|L| \leq 1$, then

$$|L| \leq |A| |\xi - p/q|.$$

When $Ap - Bq = 0$ this is equality. Otherwise the nonzero integer $Ap - Bq$ gives $1/q \leq |L| + |A||\xi - p/q|$, which proves the inequality. We fix $0 < \eta < \tau$ and use the preceding estimates in the form

$$e^{-(\tau+\eta)n^2} \leq Q_n \xi - P_n \leq e^{-(\tau-\eta)n^2}, \quad |Q_n| \leq e^{(\alpha+\eta)n^2}$$

for all sufficiently large n . The upper remainder bound determines our choice: for a sufficiently large denominator q , take $n = \lceil \sqrt{\log(2q)/(\tau - \eta)} \rceil$. Then $2q(Q_n \xi - P_n) \leq 1$. The preceding integer argument now applies to $(A, B) = (Q_n, P_n)$ for every numerator p . Also $Q_n \neq 0$: otherwise $Q_n \xi - P_n$ would be an integer strictly between 0 and 1. The lower remainder bound and the coefficient bound now give

$$|\xi - p/q| \geq e^{-(\alpha+\tau+2\eta)n^2} = q^{-(\alpha+\tau+2\eta)/(\tau-\eta)-o(1)},$$

since $n^2 = \log(2q)/(\tau - \eta) + O_\eta(\sqrt{\log q} + 1)$. The rounding error is $o(\log q)$, so it does not change the exponent. This choice covers every sufficiently large approximation denominator q ; no monotonicity assumption on the coefficients Q_n is used.

Letting $\eta \downarrow 0$, we deduce the bound $1 + \alpha/\tau = (1 - \theta)/(\theta^* - \theta)$. A common power multiplies α and τ by r , so the quotient is unchanged, although the constants in the approximation inequality may depend on r . For $31/4$, the exact interval bounds in the companion record, [Section 2.5](#), give $\theta < 0.4036982$ and $\theta^* > 0.40568$. The quotient increases with θ and decreases with θ^* in this range, so

$$\frac{1 - \theta}{\theta^* - \theta} < \frac{1 - 0.4036982}{0.40568 - 0.4036982} = \frac{2981509}{9909} < 301.$$

The same section gives the rational certificates for these bounds and sharper enclosures of the constants. $\square$

4.1 Comparison with earlier criteria

Bundschuh and Väänänen's Theorem 2 at $\alpha = -1$ [[2](#), p. 177] gives irrationality for $\log b / \log a < \theta_{\text{BV}} := 1/2 - 1/\pi^2$. Their q is the base $a/b > 1$, not its reciprocal. Their logarithmic derivative is $L_q(z) = \sum_{j \geq 1} (q^j + z)^{-1}$, so the value at $z = \alpha = -1$ is exactly $F(a/b)$. The rational height is $h(q) = a$, and their parameter $\lambda = \log h(q) / \log q$ is $1/(1 - \log b / \log a)$, which gives the displayed cutoff. Since $\pi^2 < 10$, one has $\theta_{\text{BV}} < 2/5 < \log 4 / \log 31$, so $31/4$ lies outside that sufficient region. The two sufficient regions differ on $[\theta_{\text{BV}}, \theta^*)$. Zudilin also notes an extension to noninteger rational bases $p = r/s$ for the generalized q -logarithm when $\log |r| > c \log |s|$, with $c > 0$ computable but unspecified [[5](#), Sec. 2, p. 4]. For F , the specialisation above gives $c = \mu$, the exponent bound of [[4](#), p. 162].

The positive-remainder estimates require $x > 1$, so they give no statement for negative bases. The series diverges at $x = 1$. The companion record also checks the parameters at $7/2$ in its section on Bundschuh and Väänänen's height criterion. The analytic irrationality result used there is their cited theorem.

5 Further questions

The determinant result leaves an arithmetic question: how much does clearing its coefficient denominators cost? For the elementary integer contradiction, the cleared determinants must be nonzero and tend to zero. The moment representation proves positivity, not the required denominator estimate. The [rational-value model in the companion paper](#) has the same formal orders, leading coefficients and type of fixed-base power correction, despite a rational target value. Its denominator calculation explains why these analytic properties do not produce small nonzero integer forms. The model and that calculation have ordinary proofs there and remain unformalised. The companion paper also gives an [ordinary argument for the ratio-limit extension](#) under $a_{k+1}/a_k \rightarrow \rho \in (0, q^{-1})$, without the polynomial shift bound in Theorem 2.1. That extension remains unformalised.

A different question concerns the polynomial coefficients of the approximants, rather than their remainders. Van Assche's little- q -Legendre construction [12, §3] and the multiple-orthogonality construction of Postelmans and Van Assche [13, §2] make the normalisation important. In the 2016 family, the companion paper proves positivity of the first matrix of a coefficient pencil for sizes $N \leq 8$, with real, interlacing roots below $F(p)$ for $p > 1$. The shifted minors and coefficientwise positivity are supported by the stated finite polynomial computations. The [coefficient-moment section](#) gives the definitions and the precise ranges. Positivity for every N and convergence of the largest root remain open. The general moment criteria of Wang and Zhu [15] and Sokal and Walrad [17], Berg's factorial-power example [16, Theorem 5.1], and the quadrature construction of Golub and Welsch [18] provide the relevant comparisons. Using a recurrence from another family would require a proof that it holds for these coefficients. The root-of-unity method of Krattenthaler, Rochev, Väänänen and Zudilin [14], for example, concerns a separate construction.

5.1 Congruences and divided forms

At $3/2$, a polynomial Q of degree at most W has integral homogenised value $H_W(Q) = 2^W Q(3/2)$. Congruences at powers of 2 and 3 give finite counting criteria for combinations of such values. The companion paper treats [integer rescaling](#) and [endpoint congruences and signed sums](#). It also treats the rank-one case and proves the quantitative finite-fibre estimate used below. Equal residues ensure that subtraction and division leave integer coefficients. They do not ensure a nonzero small remainder. Smith normal form counts the residue image of the evaluated lattice [6, Theorems 2.3–2.4]; a separate estimate must control the real remainders within each residue class.

For example, suppose that M integral rows (A_j, B_j) have remainders $e_j = A_j F(3/2) - B_j$. Let D be a positive integer, and suppose that the coefficient-pair subset sums $\sum_{j \in I} (A_j, B_j)$ take Q values in $(\mathbb{Z}/D\mathbb{Z})^2$. Put $T = \sum_j |e_j|$ and suppose that at most k subsets give any fixed remainder within a residue class. The least and greatest subset remainders are

$$\sum_{e_j < 0} e_j \quad \text{and} \quad \sum_{e_j > 0} e_j;$$

their difference is T . Thus T bounds the width of the range in each residue class as well. The count below combines three losses: the Q residue classes, the number of intervals needed to cover width T , and the multiplicity k of a remainder value. For each positive integer n , the inequality

$$2^M > Qk \left(\left\lfloor \frac{nT}{D} \right\rfloor + 1 \right) \quad (16)$$

gives two subsets with the same residue vector whose remainder values differ by an amount strictly between 0 and D/n in absolute value. Indeed, in each residue class divide the range of remainders into half-open intervals of width D/n , starting at its least value. At most $\lfloor nT/D \rfloor + 1$ intervals are needed, including a last interval when the greatest value is an endpoint. If each interval contained only one remainder value, it would contain at most k subsets, contrary to (16). Thus some interval contains two different remainder values; the half-open convention makes their separation strictly less than D/n . Subtracting the two subset sums and dividing their coefficients by D gives an integral linear form of nonzero absolute value less than $1/n$. The [quantitative lemma and its proof](#) state the finite-set version without assumptions about Lambert series. Distinct subsets alone would not suffice: their sums could be equal, or distinct coefficient rows could have equal remainders. The bound k controls repeated real values within a residue class, not merely repeated rows.

The elementary integer-base method provides a different comparison. Vandehey [10] and Duverney and Tachiya [11] use integer-base expansions. At a noninteger rational base the [cleared-tail recurrence](#) has a growing denominator contribution. The [clearing and tail-recurrence calculations](#) in the companion paper include the [coordinatewise obstruction at 3/2](#). They make no assertion about other approximation families. At the function level, Bell and Smertnig [9, Theorem 1.3] exclude Mahler equations for the divisor generating series, while leaving the arithmetic of a single rational argument undecided.

A construction problem. The unrestricted request for primitive polynomial rows and a signed combination whose divided remainder lies strictly between 0 and $1/n$ is equivalent to the irrationality of the target. The companion paper retains the full statement and proves this equivalence in its [section on divided linear forms](#). A more specific question must prescribe the approximation family. For the 2004 family one can instead ask whether a parameter direction improves θ^* . The [all-base degree restriction](#) proves an upper bound of $1/2$ for the degree cutoff. Its hypotheses require one polynomial family, with common leading constants in its degree, coefficient-height and remainder estimates at every fixed real base $x > 1$; the error terms may depend on x . Merely having estimates at $3/2$ does not suffice.

Under those hypotheses, the [nondecay corollary](#) goes beyond the failure of a sufficient criterion: after denominator clearing, the undivided forms cannot tend to zero at any rational base a/b with $b < a < b^2$, including $3/2$, even without a quadratic degree limit. If that limit exists, their absolute values tend to infinity in this strict region. Division by additional base-dependent content and families outside these hypotheses remain unclassified. Equality in the rational-base cutoff is also unresolved by the estimates of Section 4.

A Verification and reproducibility

The [evidence record](#) indexes the existing Lean proofs and recorded Comparator comparisons at the [source revision](#). Theorem 2.1 is the polynomial-bound specialisation of the [geometric-moment proof](#) cited in the companion paper; its Comparator association remains pending. The broader ratio-limit argument and rational-value model there have ordinary proofs only. This revision has had no fresh Lean or Comparator run and no independent review.

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator* links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The [verification record](#) gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

Data availability. The cited source revision contains the Lean sources, toolchain and library manifest. The companion paper supplies the longer algebraic arguments and the precise scope of the finite computations.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator	Theorem 2.1	Lean · Comparator pending
Proposition 3.1	Lean · Comparator	Theorem 3.2	Lean · Comparator
Theorem 4.1	Lean · Comparator	Corollary 4.2	Lean · Comparator
Corollary 4.3	Lean · Comparator		